

Background

- By 2045, statistics from the International Diabetes Federation show that approximately **783 millions**, will be living with diabetes. (International Diabetes Federation, n.d.)
- One of the most feared complications that come with diabetes mellitus is diabetic retinopathy, as it contributes to loss of quality, blurry vision, and photophobia.
- This shows that there is still a need for exploration of alternative treatments for diabetic retinopathy, as some **individuals might prefer therapeutic options** to address diabetic retinopathy.
- Drosophila melanogaster* are perfect for genetic studies as they share 75% of the genes that cause diseases in humans. With a malleable DNA code and the ability to reproduce quickly.
- Studies suggest that Rosmarinic acid helps **reduce oxidative stress and inflammation**, two of the main factors that drive diabetic retinopathy progression (Kang & Yang, 2020).
- Rosmarinic acid (RSAD) shows properties that can protect neurons from damage and regulate harmful free radicals, it shows significant potential in **preserving retinal health and slowing down vision loss** in diabetes-related complications.

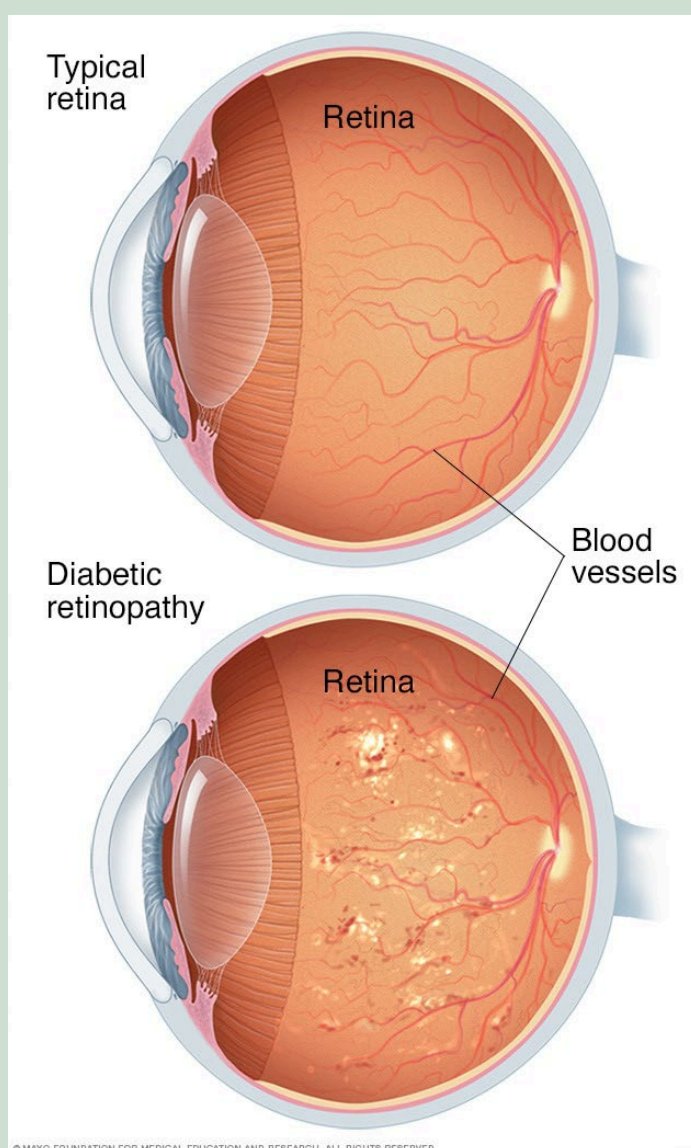


Figure 1: Infographic showing Diabetic retinopathy in the eye. Graphic from Mayo Clinic. (2023). *Diabetic Retinopathy - Symptoms and Causes*. Mayo Clinic, <https://www.mayoclinic.org>

Testing the Effects of Rosmarinic Acid from Rosemary Herb on the Visual and Neural Impairment of *Drosophila melanogaster* Induced by a High Sucrose Diet

Experimental Design

Fly Handling + Data Collection

Diets:

- Normal Diet:** Normal Diet + no RSAD
- High Sucrose Diet:** 3x the normal sucrose levels + No RSAD
- Low Rosmarinic Acid:** Normal Diet + 0.0105g RSAD
- High Rosmarinic Acid:** Normal Diet + 0.0788g RSAD
- Low Rosmarinic Acid + High Sucrose Diet:** 3x the normal sucrose levels + 0.0105g RSAD Acid
- High Rosmarinic Acid + High Sucrose Diet:** 3x the normal sucrose levels + 0.0788g RSAD Acid

Fly Care + Feeding Schedule:

- D. melanogaster* were fed their diets for 10-14 days. As soon as the 10-14 days were done, trials were run and data was collected.
- Change the fly food every 7 days.

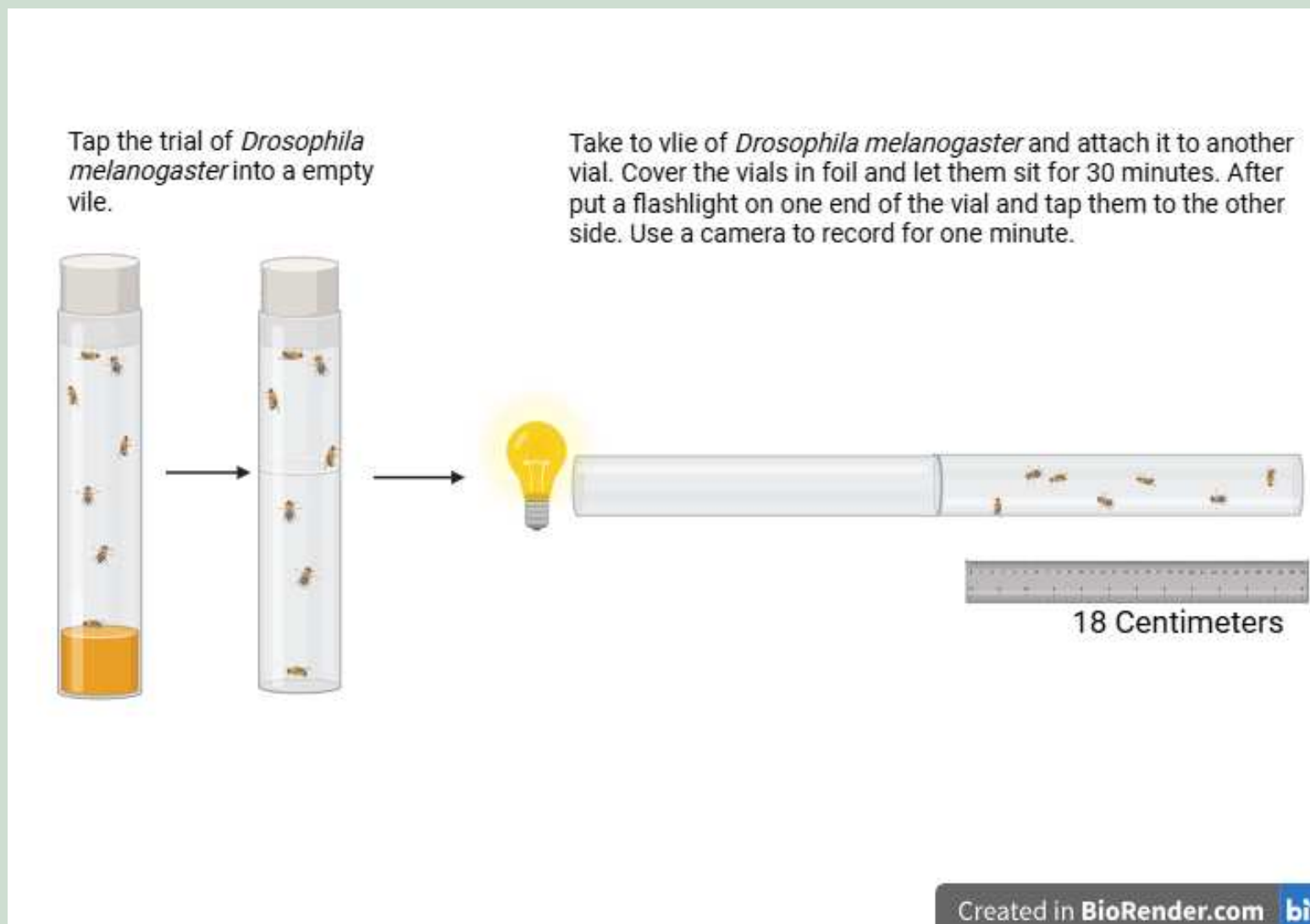


Figure 2: Infographic showing a phototaxis assay. Graphic created by the finalist using BioRender.

Phototaxis assay :

- Transfer the *D. melanogaster* into the apparatus (two vials connected).
- Set the test group into a dark environment for 30 minutes.
- After, set up a flashlight on one end of the vial and tap the *D. melanogaster* down to the other side.
- Start the camera and record the time each *D. melanogaster* takes to travel 18 centimeters.

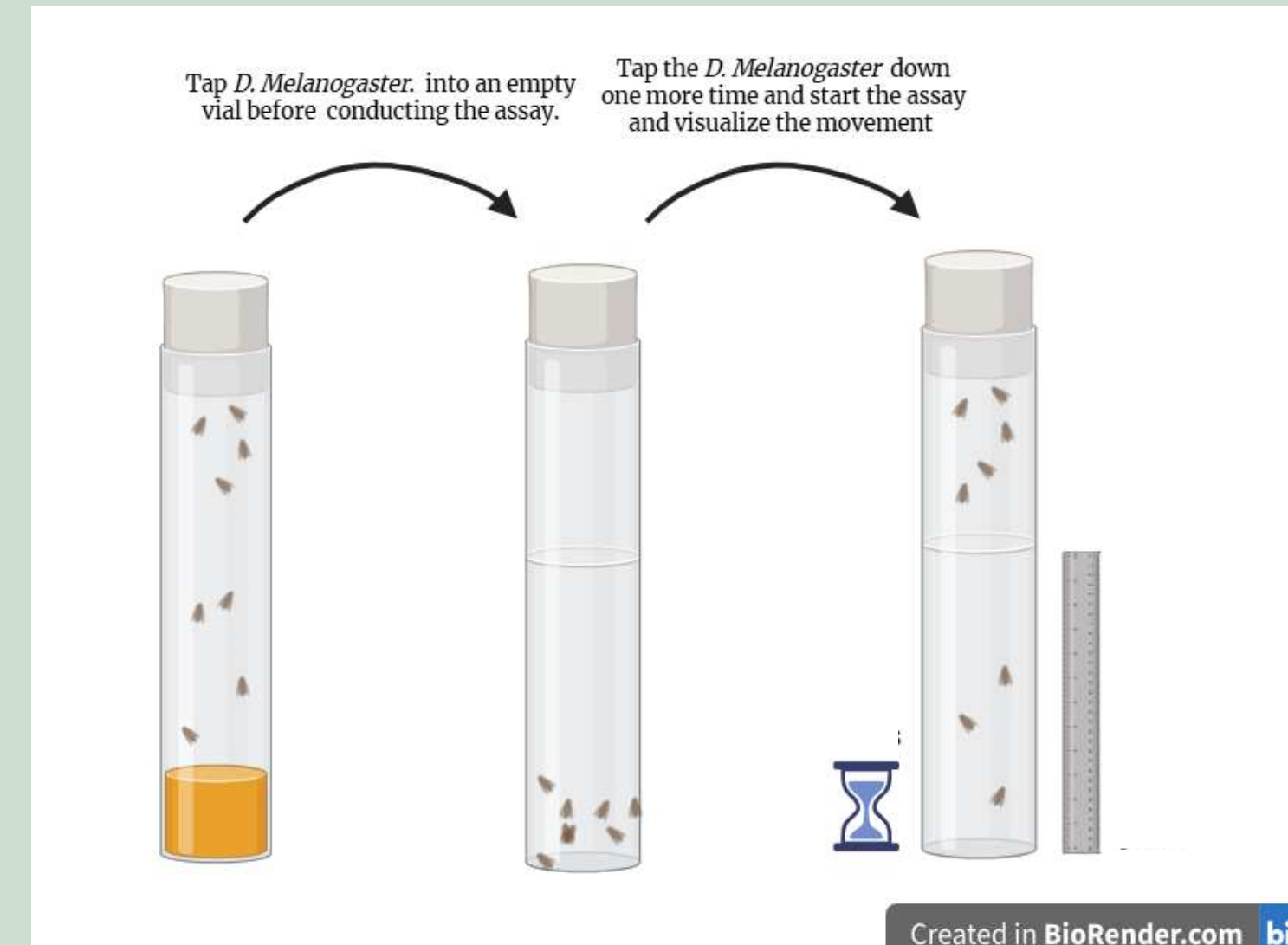


Figure 3: Infographic showing a climbing assay. Graphic created by the finalist using BioRender.

Climbing assay :

- Transfer the *D. melanogaster* into the bottom of a clean vial and place the vial into the apparatus.
- Set up a camera to capture the entire length of the vial at a fixed distance.
- Tap the apparatus three times to tap the *D. melanogaster* to the bottom of the vial.
- Start the camera and a separate timer simultaneously. Record the *D. melanogaster* climbing behavior for at least 20 seconds. Passing is considered if the *D. melanogaster* finishes the distance within 15 seconds.

Data + Visualizations

Pass and Fail Rates Across All Experimental Groups (9 cm)

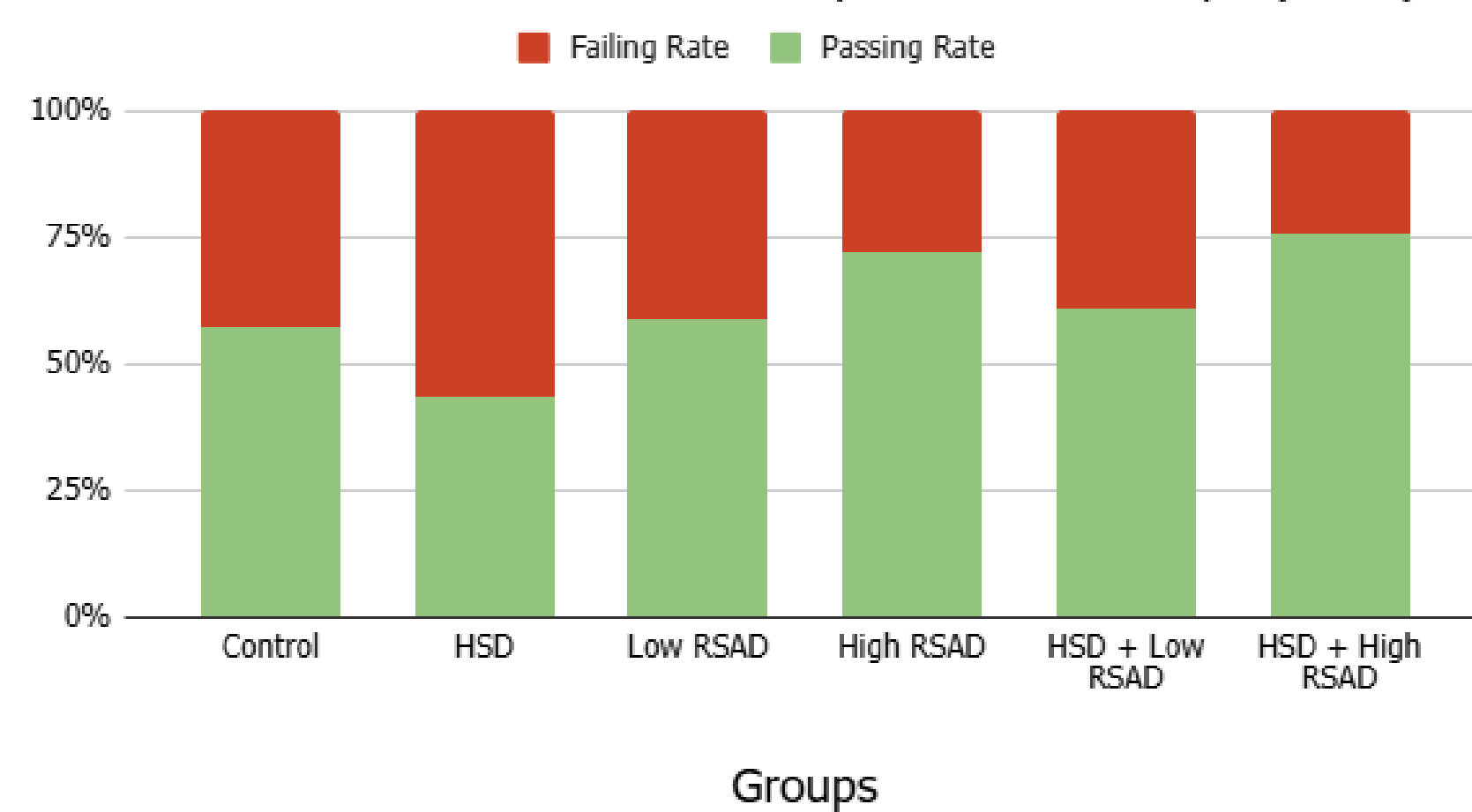


Figure 4: Passing and failing rates of *D. melanogaster* during a climbing assay. Graphic created by the finalist using Data Classroom
Key Takeaway: HSD has the biggest failure rate. High RSAD, especially when combined with the HSD, resulted in the highest climbing success rates.

Table 1: Passing rates of *D. melanogaster* during a climbing assay.

Group	Control	HSD	Low RSAD	High RSAD	Low RSAD + HSD	High RSAD + HSD
Passing Rate	58%	43%	59%	72%	61%	76%

Table 2: Passing rates of *D. melanogaster* during a phototaxis assay.

Group	Control	HSD	Low RSAD	High RSAD	Low RSAD + HSD	High RSAD + HSD
Passing Rate	90%	59%	86%	83%	76%	63%

Climbing Time (secs) Across the Different Groups (9cm)

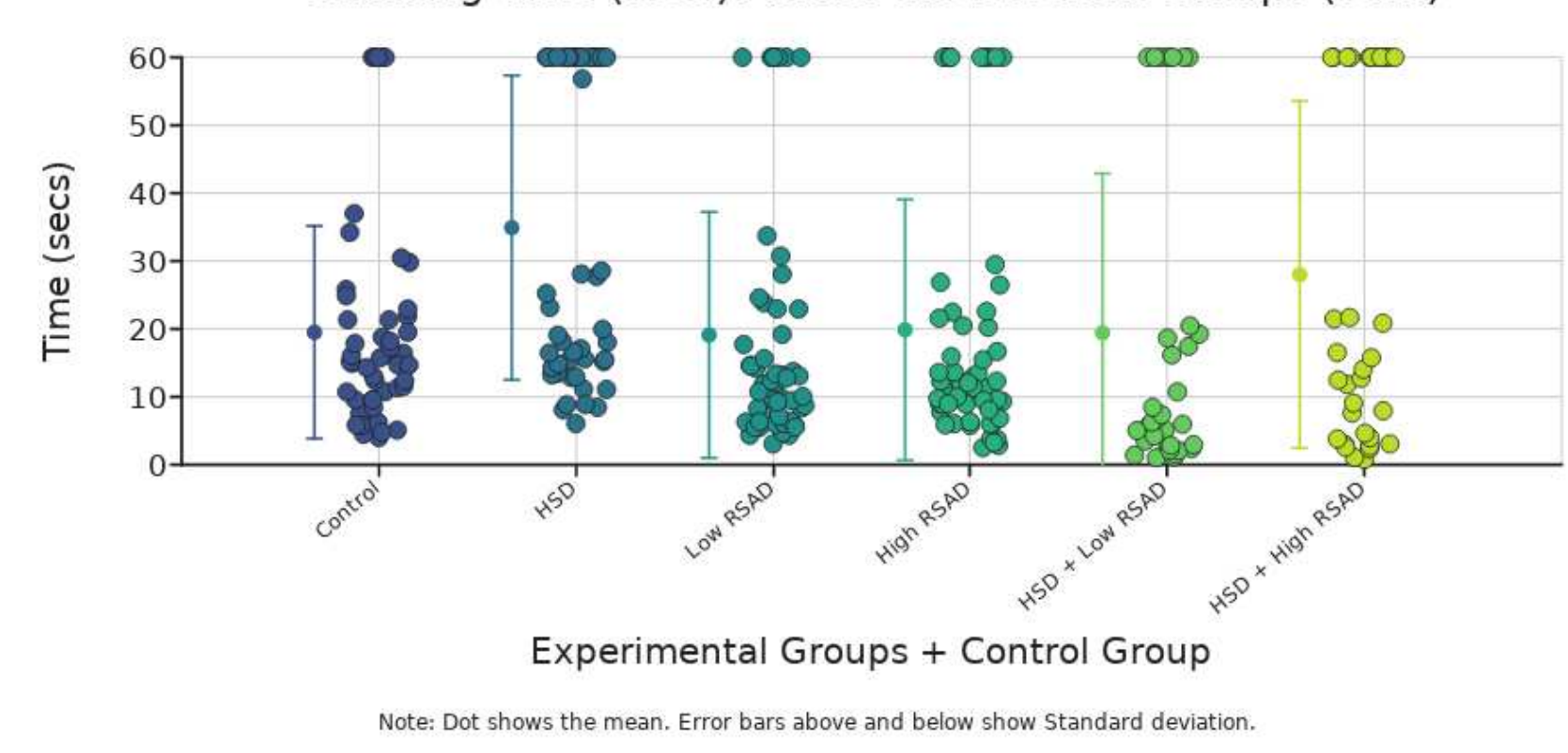


Figure 5 : Times of *D. melanogaster* during a phototaxis assay. Graphic created by the finalist using Data Classroom
Key Takeaway: HSD flies took the longest to complete the assay, suggesting a longer response time. Compared to the control and the RSAD groups.

Climbing Time (secs) Across Different Groups Excluding DNF

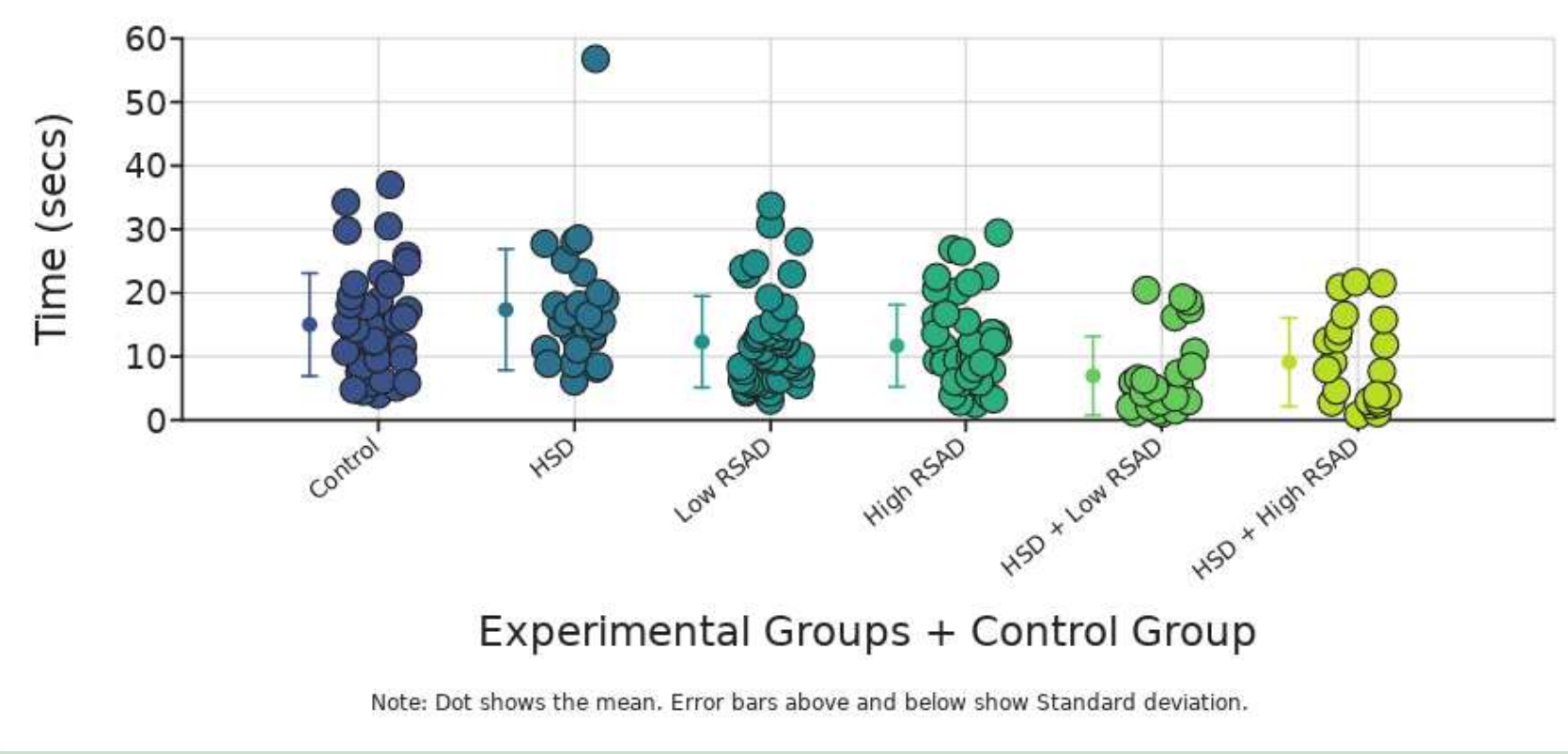


Figure 6 : Times of *D. melanogaster* during a phototaxis assay excluding the *D. melanogaster* that didn't finish. Graphic created by the finalist using Data Classroom
Key Takeaway: Even after excluding the DNF, RSAD group performed faster than HSD, reinforcing the potential protective effect of RSAD.

Results + Conclusions

The ANOVA test produced a p-value of <0.01 which shows there is a **strong confidence that there is a difference between the groups**. After running the Post Hoc Tukey's Range Test, One major finding was a strong confidence that there is a difference between the control group and the high sucrose diet group.

This means that there are 3 significant key takeaways:

- This means that the high sucrose diet **induced visual and neural impairment**. There was, on average, almost a 15% slower completion from the HSD group than the control group.
- There was also a strong confidence that the groups were not different between the Rosmarinic acid + High sucrose diets and the control groups. This showed that Rosmarinic acid **did not** have a **negative impact on their health**.
- Finally, the last finding was a strong confidence that there is a difference between the Rosmarinic acid + High sucrose diets and the high sucrose diet flies. This shows that there was an **improvement of neural and visual health**. There was on average almost a 12 second increase in time.

The Chi squared test showed with a p-value of <0.01 that there is an extremely high confidence that there is a **relationship between the groups**. This indicated that the motor performance of the flies differed through the treatments. *D. Melanogaster* in the HSD had a lower proportion of successful climbs compared to controls, which further helps prove that **high sucrose intake negatively affected the motor function**. Groups that received the RSAD showed higher passing rates than HSD, suggesting that **RSAD can be associated with positively impacting the locomotor function under HSD**.

Future work:

Results support the idea that Rosmarinic acid can be used as a treatment to combat visual and neural impairment. Though more research needs to be done to further support these findings, the results support that Rosmarinic acid mitigates high-sucrose-associated impairment in *D. melanogaster*. In the future sample sizes should be increased and creating a dose-response cure could all help create a stronger relationship between the Rosmarinic acid and visual and neural impairment.

References

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